DocuLens Test Document A

Computers, data, algorithms, and data structures

Purpose: Synthetic content for testing upload, extraction, search, highlighting, citations, comparison, tables, and export.

Note: Examples are fictional and are not professional advice.

1. Computers and information systems

A **computer** is an electronic system that accepts input, processes data according to instructions, stores information, and produces output. A computer system may include a processor, memory, storage, input devices, output devices, and software.

In this document, **data** means recorded values or observations. **Information** means data interpreted in a context. For example, 42 is data; “42 items remain in stock” is information.

An **algorithm** is a finite, ordered set of steps for solving a problem or completing a calculation. An algorithm can be described in plain language, pseudocode, or a flowchart.

2. A simple algorithm

Input: a non-empty list of numbers. Treat the first number as the current largest. Examine each remaining number. If a number is greater, replace the current largest. Return the largest after examining the list.

The procedure assumes the input list contains at least one value. An empty list requires a separate error or result policy.

Example: find the largest value

3. Data structures

An **array** stores elements in an indexed sequence. A **linked list** stores elements in nodes connected by references. A **hash table** uses a hash function to help locate values by key.

A data structure organizes data so operations such as searching, insertion, deletion, or traversal can be performed. Different structures make different trade-offs.

4. Example measurements

Illustrative table — not benchmark data

Structure	Lookup description	Insertion consideration	Example use
Array	Fast indexed access	May require shifting elements	Ordered records
Linked list	Sequential traversal in a basic implementation	Can link a node when its position is known	Frequent local insertions
Hash table	Often fast average key lookup	Collisions and resizing matter	Key-value records
Tree	Depends on tree shape and balancing	May require restructuring	Hierarchical or ordered data

Actual performance depends on implementation, data size, workload, and hardware.

5. Search test phrases

The computer stores data. The algorithm processes data. A computer can execute an algorithm, but computer, data, and algorithm do not mean the same thing.

Exact phrase: **finite, ordered set of steps**. Case test: COMPUTER, Computer, computer. Punctuation test: computer—data; algorithm/data; (computer).

Repeated sentence: The algorithm processes data. The algorithm processes data.

6. Limitations and edge cases

A procedure given an empty list cannot return the largest list element because no element exists. It may return an error, an optional value, or a documented sentinel.

The phrase “fast lookup” is incomplete unless the data structure, operation, workload, and assumptions are specified. Average-case performance is not a guarantee for every input.

Keyword trap: computer is mentioned here, but this paragraph does not define the word computer.

7. CSV export test data

Record	Label	Value	Note
R-01	Computer	42	ordinary value
R-02	Algorithm	A > B	comparison symbols
R-03	Data	=1+1	formula-injection test string
R-04	Quoted phrase	"hello, world"	commas and quotes
R-05	Unicode	café / naïve / Δ	Unicode characters

The =1+1 value is intentionally untrusted text for safe CSV export testing.

8. Evidence and citation checks

Passage A: “An algorithm is a finite, ordered set of steps for solving a problem or completing a calculation.” Use this to test exact quotation matching and page citations.

Passage B: “A data structure organizes data so operations such as searching, insertion, deletion, or traversal can be performed.”

A citation must refer to the correct file and page. If a quotation is altered, invented, or assigned to another page, it must not be labelled verified.

9. Test questions

Q1. How does this document define an algorithm?

Q2. What is the difference between data and information?

Q3. Name three data structures and a trade-off.

Q4. What assumption does the largest-value procedure make?

Q5. Does this document claim a hash table always has constant-time lookup? Answer with evidence.

Q6. Find every occurrence of computer and algorithm, then inspect each result in the reader.